

AR Expeditions

{newworldsreading}

AR Expeditions explores the use of artificial intelligence (AI) throughout the full development cycle of an augmented reality (AR) learning tool designed to engage students in grades 3–5 and their families. This project integrates interactive story-based AR experiences to promote meaningful family engagement and spark curiosity in shared learning.

A COLLABORATION BETWEEN:



ADVANCING EDUCATIONAL INNOVATION THROUGH AI-ENHANCED DESIGN AND IMMERSIVE AR ENVIRONMENTS

Ideation & Brainstorming

This stage focuses on generating and refining creative concepts for the AR experiences. Using AI tools, the team explored story ideas, visual directions, and early prototypes to establish the foundation for development.

AI Toolkit:

CONCEPT ART

COLLABORATION

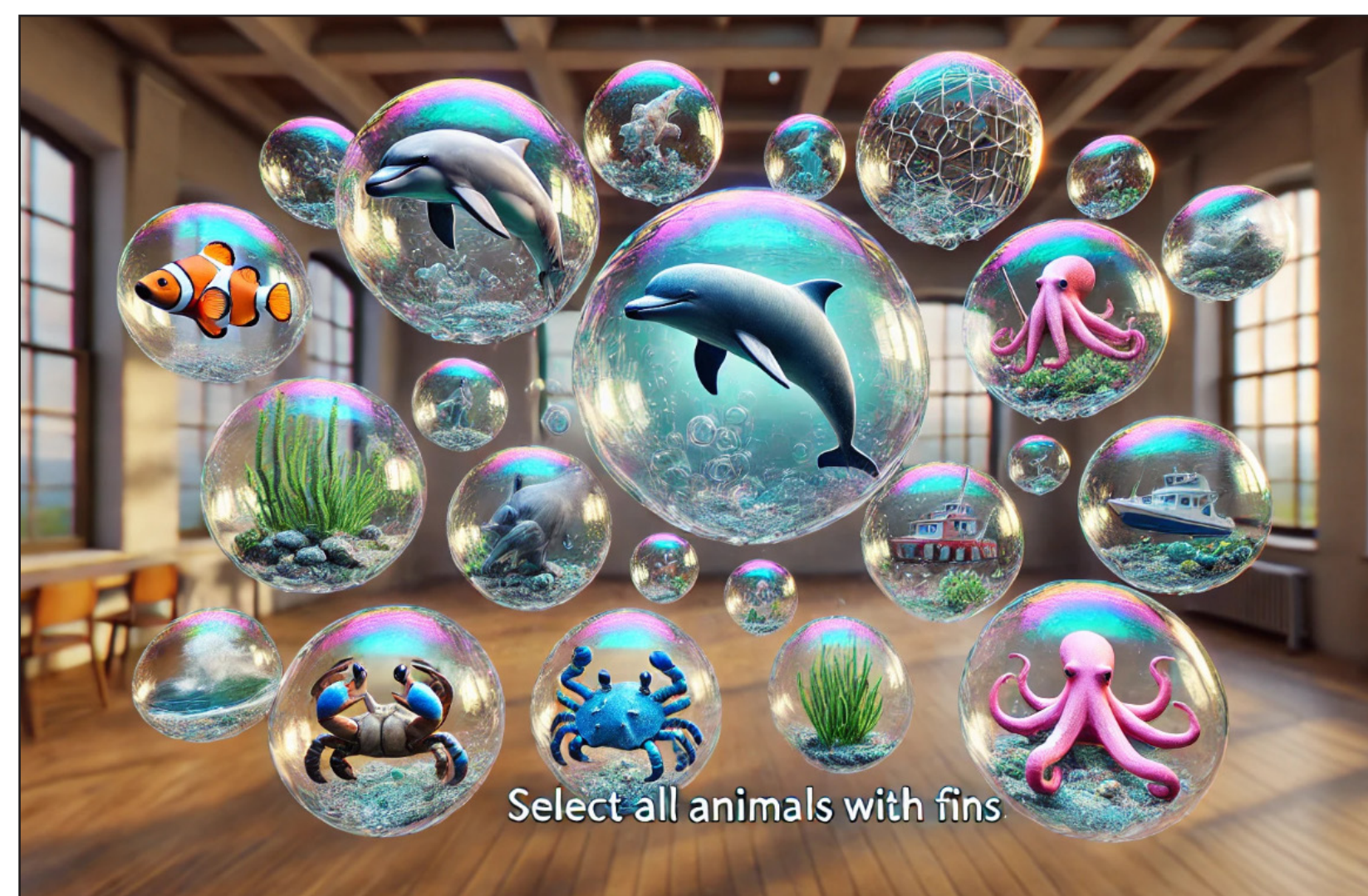


Challenges:

- Generating, organizing, and refining large volumes of creative ideas
- Balancing a wide breadth of topics and curriculum-aligned content
- Managing scope and feasibility across creative and technical teams
- Developing interactive prototypes, concept art, and 3D models for early-stage visualization.
- Translating concepts for a 3D AR space while working on 2D interfaces

AI-Enhanced Solutions:

- Accelerated ideation by generating multiple concept directions, themes, and narrative possibilities rapidly
- Enhanced visualization through AI-generated concept art that served as a visual reference for designers and modelers
- Facilitated collaborative refinement by clustering, chunking, and iterating on visually diverse ideas within Miro AI
- Bridged communication gaps by simplifying technical jargon between creative, instructional, and development teams
- Produced a wide range of concept art using ChatGPT 4o and Gemini Imagen to explore variations quickly and efficiently
- Refined and polished AI-generated visuals with Adobe Firefly to ensure clarity and professional quality



AR Gameplay & User Experience

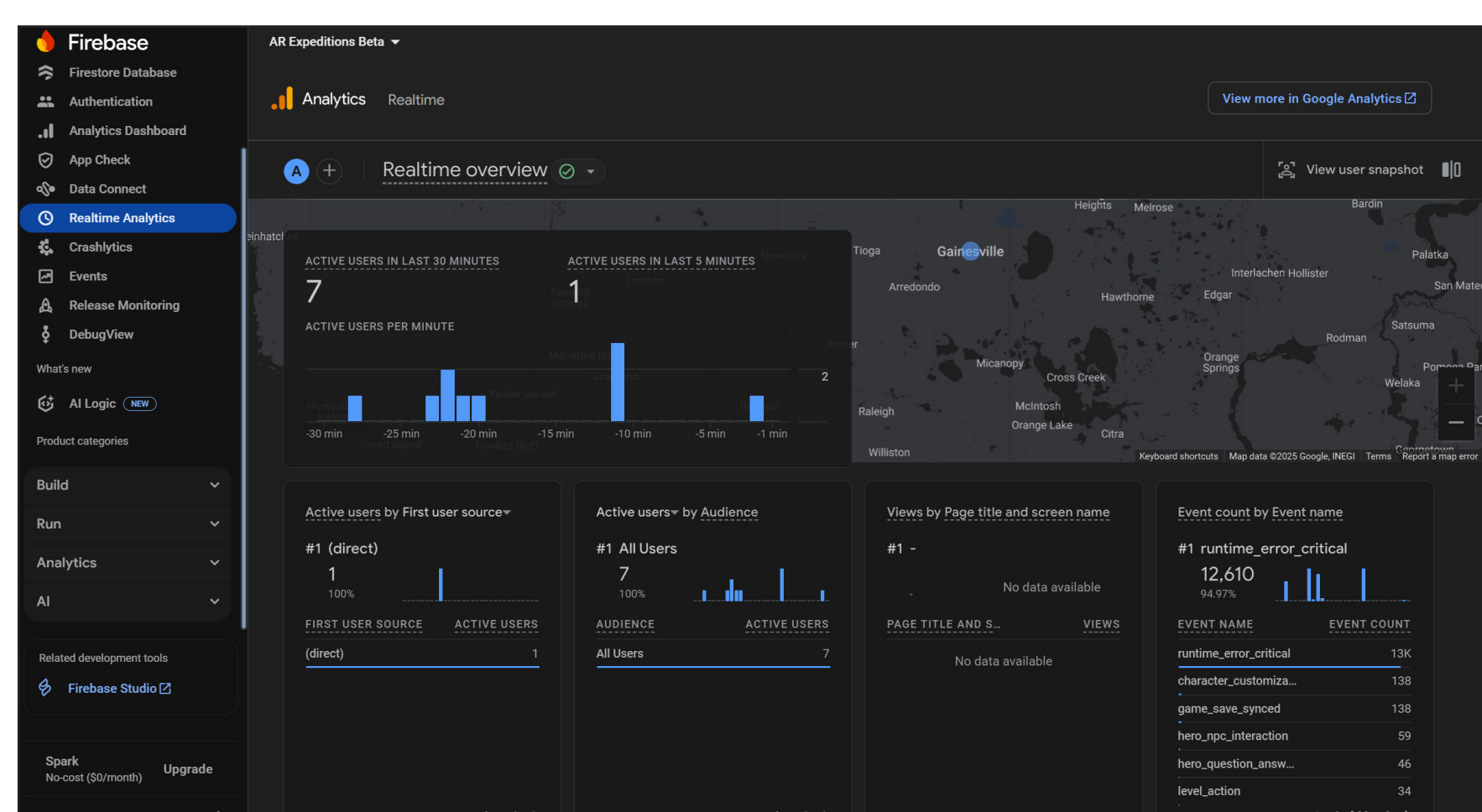
This phase focuses on player interaction with augmented reality (AR), accessibility, and educational engagement. AI tools analyzed gameplay data to inform adaptive mechanics and data-driven refinements that enhanced the user experience for future experiences and builds.

Challenges:

- Ensuring smooth placement, interaction, and tracking of digital objects in real-world AR environments.
- Designing dynamic, interactive gameplay where AI-controlled characters react to players and the environment.

AI-Enhanced Solutions:

- Analyzed complex gameplay data to generate summaries and action plans, identifying errors and opportunities to improve player engagement and learning outcomes
- Prototyped an AI-driven analytics dashboard with Loveable to visualize user behavior and inform future gameplay refinements
- Utilized Unity ML Agents and OPSIVE Behavior Trees to define decision-making pathways, enabling AI characters to respond dynamically to player interactions and environmental changes
- Applied ARCore and ARKit for real-time surface detection and object anchoring, ensuring consistent and seamless AR integration across devices



AI Toolkit:

DATA ANALYTICS & TELEMETRY



ANALYTICS DASHBOARD

BEHAVIOR SYSTEMS



Content Development

This phase centers on building the narrative, dialogue, and auditory experience. AI supported the creation of age-appropriate language, literary elements, and immersive soundscapes that align with New Worlds Reading content.

Challenges:

- Crafting developmentally appropriate dialogue aligned with literacy standards and New Worlds Reading books
- Ensuring consistency in tone and reading level across characters and experiences
- Synthesizing qualitative user feedback into actionable content revisions
- Creating immersive soundscapes that enhance narrative and gameplay without overwhelming learners
- Producing custom audio effects, background music, and voiceovers for language content

AI Toolkit:

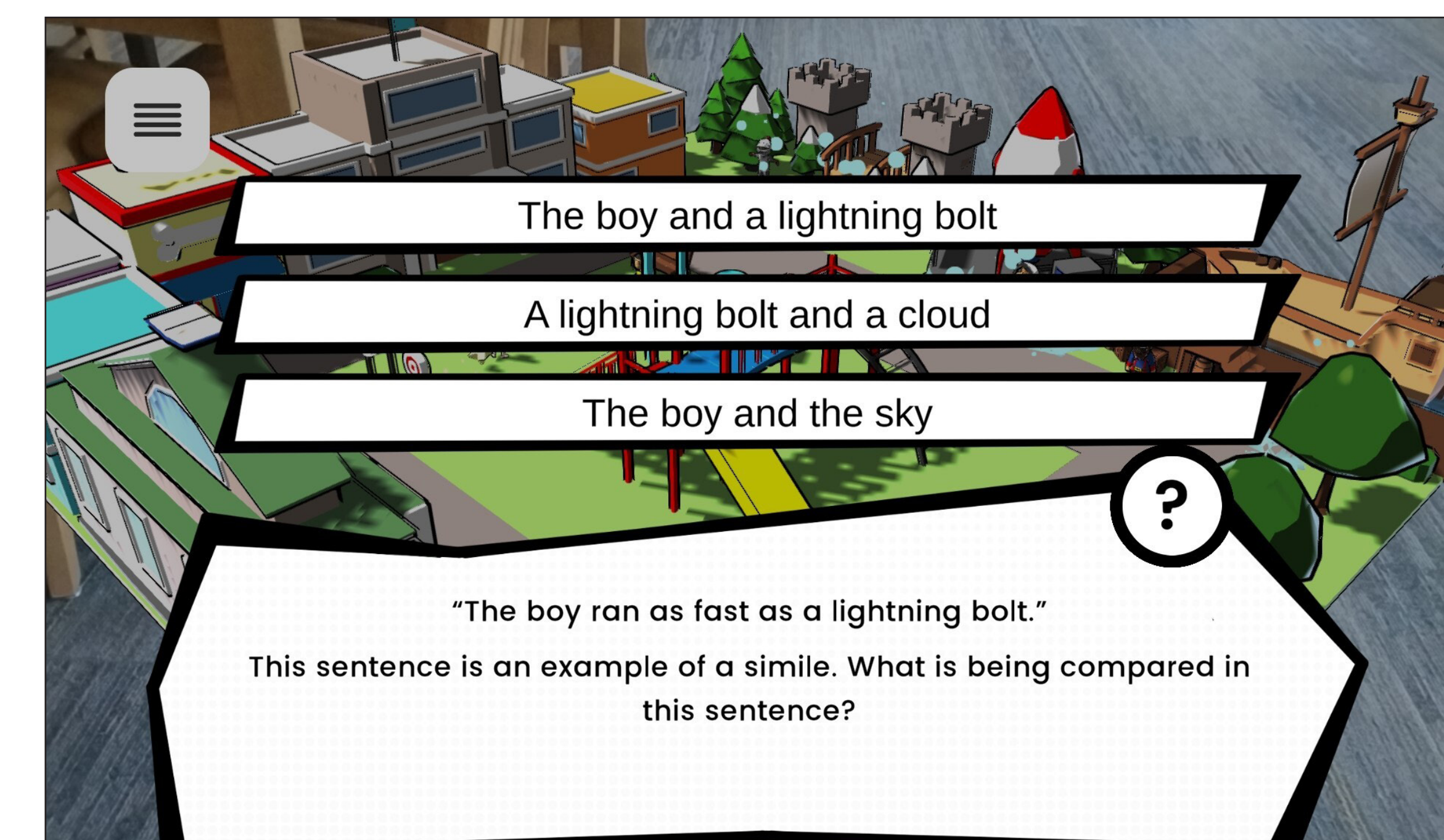
NARRATIVE DESIGN

AUDIO & MUSIC



AI-Enhanced Solutions:

- Generated and refined dialogue drafts using ChatGPT, tailored to specific reading levels, emotional tone, and character personality
- Summarized and analyzed user feedback through ChatGPT to identify patterns in comprehension and engagement for iterative content improvement
- Produced looping instrumental tracks, ambient soundscapes, and thematic music with SUNO to match gameplay environments (e.g., underwater scenes)
- Created high-quality, natural voiceovers with Eleven Labs (Deep Voice for Unity), supporting bimodal reading and accessibility



Technical Implementation

This stage focuses on coding, asset integration, and maintaining stability across devices and AR systems. AI tools streamlined development, debugging, and animation to speed production and boost performance.

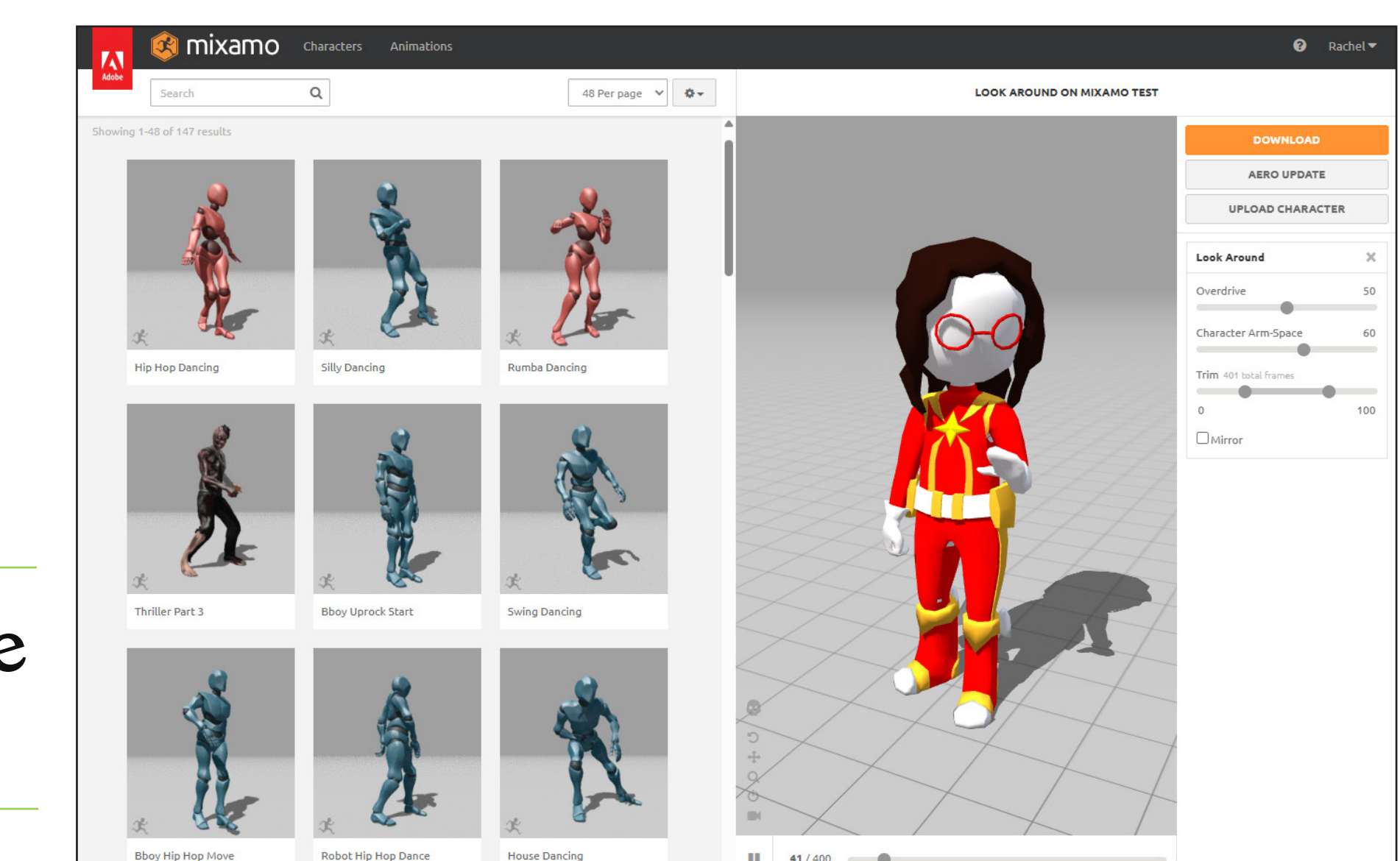
AI Toolkit:

CODE AGENT



AR PLANE DETECTION

ANIMATION



Challenges:

- Implementing multiple game levels and managing scene transitions, and persistent data
- Programming complex mechanics including physics, AI interactions, and animations while maintaining optimized performance
- Debugging existing scripts to ensure functionality and prevent logic conflicts
- Integrating a wide range of libraries and frameworks (ARKit, ARCore, Unity UI, third-party SDKs) without compatibility issues
- Optimizing performance across diverse mobile devices with varying processing power and camera capabilities
- Maintaining accurate AR tracking, lighting estimation, and anchor persistence in dynamic real-world settings

AI-Enhanced Solutions:

- Generated and refined Unity C# scripts using NaviGator AI and Claude, accelerating the development of gameplay systems and interaction logic
- Provided real-time code completion, syntax correction, and debugging support through GitHub Copilot, reducing manual troubleshooting time
- Interpreted screen captures of in-progress builds via GPT-4 Vision to identify and resolve script or UI issues efficiently
- Automated character rigging and animation blending using Adobe Mixamo's AI-driven systems, ensuring consistent and natural movement
- Optimized for performance, battery efficiency, and AR stability through AI-assisted code reviews and iterative testing